

1) Menu-driven program for CREATE, DISPLAY, INSERT, DELETE and LINEAR SEARCH

```
#include <iostream>
using namespace std;

int main()
{
    int a[100], n = 0, choice, pos, value, key, i;

    while (1)
    {
        cout << "\n----- MENU ----- \n";
        cout << "1. CREATE\n";
        cout << "2. DISPLAY\n";
        cout << "3. INSERT\n";
        cout << "4. DELETE\n";
        cout << "5. LINEAR SEARCH\n";
        cout << "6. EXIT\n";
        cout << "Enter choice: ";
        cin >> choice;

        if (choice == 1)
        {
            cout << "Enter number of elements: ";
            cin >> n;
            cout << "Enter elements:\n";
            for (i = 0; i < n; i++)
                cin >> a[i];
        }
        else if (choice == 2)
        {
            cout << "Array: ";
            for (i = 0; i < n; i++)
                cout << a[i] << " ";
            cout << "\n";
        }
        else if (choice == 3)
        {
            cout << "Enter position (1 to " << n + 1 << "): ";
            cin >> pos;
            cout << "Enter value: ";
            cin >> value;

            if (pos >= 1 && pos <= n + 1)
            {
                for (i = n; i >= pos; i--)
                    a[i] = a[i - 1];
                a[pos - 1] = value;
                n++;
            }
            else
                cout << "Invalid position\n";
        }
        else if (choice == 4)
        {
            cout << "Enter position to delete: ";
            cin >> pos;

            if (pos >= 1 && pos <= n)
            {
                for (i = pos - 1; i < n - 1; i++)
                    a[i] = a[i + 1];
                n--;
            }
            else
                cout << "Invalid position\n";
        }
        else if (choice == 5)
        {

```

```

        cout << "Enter element to search: ";
        cin >> key;

        for (i = 0; i < n; i++)
        {
            if (a[i] == key)
            {
                cout << "Element found at position " << i + 1 << "\n";
                break;
            }
        }

        if (i == n)
            cout << "Element not found\n";
    }
    else if (choice == 6)
    {
        break;
    }
    else
    {
        cout << "Invalid choice\n";
    }
}

return 0;
}

```

2) Remove duplicate elements and keep only unique elements

```

#include <iostream>
using namespace std;

int main()
{
    int a[100], n, i, j, k;

    cout << "Enter number of elements: ";
    cin >> n;

    cout << "Enter elements:\n";
    for (i = 0; i < n; i++)
        cin >> a[i];

    for (i = 0; i < n; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (a[i] == a[j])
            {
                for (k = j; k < n - 1; k++)
                    a[k] = a[k + 1];
                n--;
                j--;
            }
        }
    }

    cout << "Array after removing duplicates:\n";
    for (i = 0; i < n; i++)
        cout << a[i] << " ";

    return 0;
}

```

3) Rotate a 1-D array by K positions (left and right rotation)

```

#include <iostream>
using namespace std;

int main()
{
    int a[100], n, k, i, j, temp;

    cout << "Enter number of elements: ";
}

```

```

cin >> n;

cout << "Enter elements:\n";
for (i = 0; i < n; i++)
    cin >> a[i];

cout << "Enter K: ";
cin >> k;

k = k % n;

// Left rotation
for (i = 0; i < k; i++)
{
    temp = a[0];
    for (j = 0; j < n - 1; j++)
        a[j] = a[j + 1];
    a[n - 1] = temp;
}

cout << "Left rotation: ";
for (i = 0; i < n; i++)
    cout << a[i] << " ";

// Right rotation
// Enter the original array again for a separate right rotation.
cout << "\nEnter the original array again for right rotation:\n";
for (i = 0; i < n; i++)
    cin >> a[i];

for (i = 0; i < k; i++)
{
    temp = a[n - 1];
    for (j = n - 1; j > 0; j--)
        a[j] = a[j - 1];
    a[0] = temp;
}

cout << "Right rotation: ";
for (i = 0; i < n; i++)
    cout << a[i] << " ";

return 0;
}

```

4(a) Reverse the elements of an array

```

#include <iostream>
using namespace std;

int main()
{
    int a[100], n, i, temp;

    cin >> n;
    for (i = 0; i < n; i++)
        cin >> a[i];

    for (i = 0; i < n / 2; i++)
    {
        temp = a[i];
        a[i] = a[n - 1 - i];
        a[n - 1 - i] = temp;
    }

    for (i = 0; i < n; i++)
        cout << a[i] << " ";

    return 0;
}

```

4(b) Matrix multiplication

```

#include <iostream>
using namespace std;

```

```

int main()
{
    int a[10][10], b[10][10], c[10][10];
    int r1, c1, r2, c2, i, j, k;

    cout << "Enter rows and columns of first matrix: ";
    cin >> r1 >> c1;

    cout << "Enter first matrix:\n";
    for (i = 0; i < r1; i++)
        for (j = 0; j < c1; j++)
            cin >> a[i][j];

    cout << "Enter rows and columns of second matrix: ";
    cin >> r2 >> c2;

    if (c1 != r2)
    {
        cout << "Matrix multiplication not possible";
        return 0;
    }

    cout << "Enter second matrix:\n";
    for (i = 0; i < r2; i++)
        for (j = 0; j < c2; j++)
            cin >> b[i][j];

    for (i = 0; i < r1; i++)
    {
        for (j = 0; j < c2; j++)
        {
            c[i][j] = 0;
            for (k = 0; k < c1; k++)
                c[i][j] += a[i][k] * b[k][j];
        }
    }

    cout << "Result:\n";
    for (i = 0; i < r1; i++)
    {
        for (j = 0; j < c2; j++)
            cout << c[i][j] << " ";
        cout << "\n";
    }

    return 0;
}

```

4(c) Transpose of a matrix

```

#include <iostream>
using namespace std;

int main()
{
    int a[10][10], t[10][10];
    int r, c, i, j;

    cout << "Enter rows and columns: ";
    cin >> r >> c;

    cout << "Enter matrix:\n";
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            cin >> a[i][j];

    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            t[j][i] = a[i][j];

    cout << "Transpose:\n";
    for (i = 0; i < c; i++)
    {
        for (j = 0; j < r; j++)
            cout << t[i][j] << " ";
        cout << "\n";
    }
}

```

```
    return 0;
}
```

5) Sum of every row and every column of a two-dimensional array

```
#include <iostream>
using namespace std;

int main()
{
    int a[10][10], r, c, i, j;
    int sum;

    cout << "Enter rows and columns: ";
    cin >> r >> c;

    cout << "Enter matrix:\n";
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            cin >> a[i][j];

    for (i = 0; i < r; i++)
    {
        sum = 0;
        for (j = 0; j < c; j++)
            sum += a[i][j];
        cout << "Sum of row " << i + 1 << " = " << sum << "\n";
    }

    for (j = 0; j < c; j++)
    {
        sum = 0;
        for (i = 0; i < r; i++)
            sum += a[i][j];
        cout << "Sum of column " << j + 1 << " = " << sum << "\n";
    }

    return 0;
}
```